(a) Mg^{2+} (b) Ti^{3+} (c) V^{3+} (d) Fe^{2+}

6. The statement that is not correct for the periodic classification of element is **[1992 - 1 Mark]**

(a) The properties of elements are the periodic functions of their atomic numbers

(b) Non-metallic elements are lesser in number than metallic elements

(c) The first ionisation energies of elements along a period do not vary in a regular manner with increase in atomic number

(d) For transition elements the *d*-subshells are filled with electrons monotonically with increase in atomic number.

3 Numeric / New Stem Based Questions

7. If IUPAC name of an element is "Unununnium" then the element belongs to n^{th} group of Periodic table. The value of n is ___11___. **[Main Jan. 30, 2024 (I)]**

6 MCQs with One or More than One Correct Answer

8. The statements that **are true** for the long form of the periodic table are : **[1988 - 1 Mark]**

(a) it reflects the sequence of filling the electrons in the order of sub-energy level s, p, d and f.

(b) it helps to predict the stable valency states of the elements

(c) it reflects trends in physical and chemical properties of the elements

(d) it helps to predict the relative ionicity of the bond between any two elements.

2. The electron affinity value are negative for :

[Main April 6, 2024 (I)]

A. $Be \rightarrow Be^-$; B. $N \rightarrow N^-$; C. $O \rightarrow O_2^-$; D. $Na \rightarrow Na^-$

E. $Al \rightarrow Al^-$

Choose the most appropriate answer from the options given below :

(a) D and E only
(b) A, B, D and E only
(c) A and D only
(d) A, B and C only

3. Evaluate the following statements related to group 14 elements for their correctness. [Main April 6, 2024 (II)]

(A) Covalent radius decreases down the group from C to Pb in a regular manner.

(B) Electronegativity decreases from C to Pb down the group gradually.

(C) Maximum covalence of C is 4 whereas other elements can expand their covalence due to presence of d orbitals.

(D) Heavier elements do not form $p\pi$-$p\pi$ bonds.

(E) Carbon can exhibit negative oxidation states.

Choose the **correct** answer from the options given below:

(a) (C), (D) and (E) Only
(b) (A) and (B) Only
(c) (A), (B) and (C) Only
(d) (C) and (D) Only

13. Among the statements (I - IV), the **correct** ones are :

[Main Sep. 03, 2020 (II)]

(I) Be has smaller atomic radius compared to Mg.

(II) Be has higher ionization enthalpy than Al.

(III) Charge/radius ratio of Be is greater than that of Al.

(IV) Both Be and Al form mainly covalent compounds.

(a) (I), (II) and (IV)
(b) (I), (III) and (IV)
(c) (II), (III) and (IV)
(d) (I), (II) and (III)

54. Within each pair of elements F & Cl, S & Se, and Li & Na, respectively, the elements that release more energy upon an electron gain are: **[Main Jan. 07, 2020 (II)]**

(a) Cl, Se and Na (b) Cl, S and Li

(c) F, S and Li (d) F, Se and Na

(c) $Na < Li < K$ (d) $Fe^{3+} < Fe^{2+} < Fe^{4+}$

75 The correct order of acidic strength is **[2000S]**

(a) $Cl_2O_7 > SO_2 > P_4O_{10}$ (b) $CO_2 > N_2O_5 > SO_3$

(c) $Na_2O > MgO > Al_2O_3$ (d) $K_2O > CaO > MgO$

[Main Feb. 1, 2024 (II)]

106. **Statements – I:** Along the period, the chemical reactivity of the elements gradually increases from group 1 to group 18.

Statement – II: The nature of oxides formed by group 1 elements is basic while that of group 17 elements is acidic.

[Main Jan. 30, 2024 (II)]

107. **Statement I :** The electronegativity of group 14 elements from Si to Pb, gradually decreases.

Statement II : Group 14 contains non-metallic, metallic, as well as metalloid elements. [Main Jan. 29, 2024 (I)]

102. Match List-I with List-II. [Main June 28, 2022 (II)]

List-I (Oxide)	List-II (Nature)
(A) Cl_2O_7	(I) Amphoteric
(B) Na_2O	(II) Basic
(C) Al_2O_3	(III) Neutral
(D) N_2O	(IV) Acidic

Choose the correct answer from the options given below:

(a) (A)–(IV), (B)–(III), (C)–(I), (D)–(II)

(b) (A)–(IV), (B)–(II), (C)–(I), (D)–(III)

(c) (A)–(II), (B)–(IV), (C)–(III), (D)–(I)

(d) (A)–(I), (B)–(II), (C)–(III), (D)–(IV)